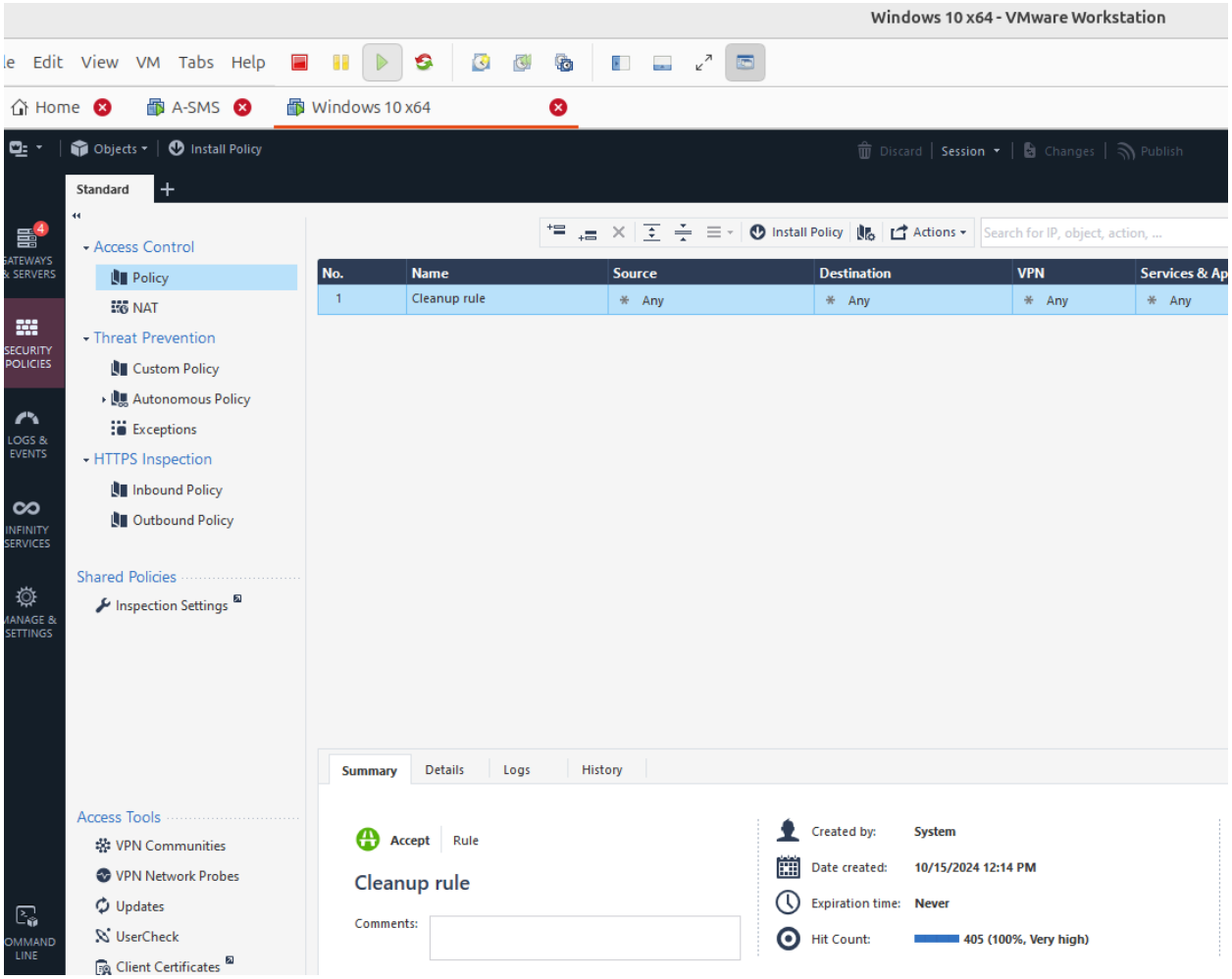
NAME: JIBRIL DAUDA MUHAMMAD
STUDENT ID: C11/26/CCSA/17109
DATE: 4th NOVEMBER, 2026

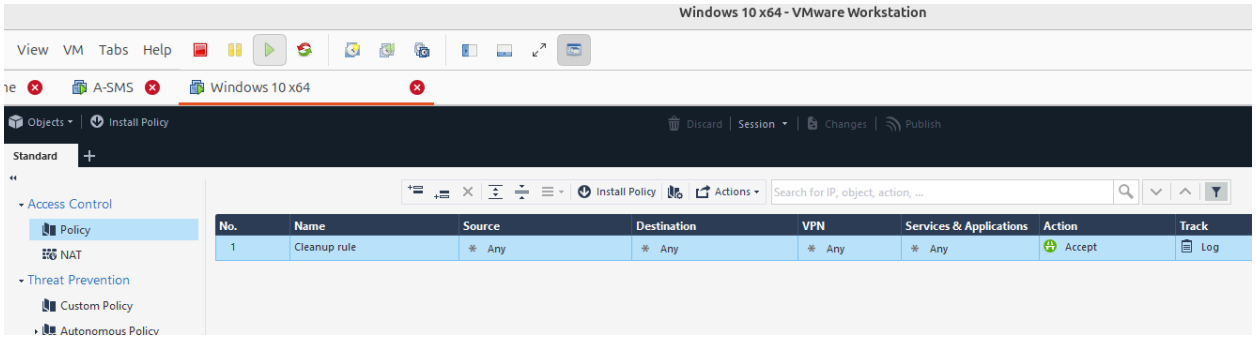
PART A

The existing Access Control policy was inspected in Check Point SmartConsole to understand its structure and default security behaviour. The Cleanup Rule was identified at the bottom of the rulebase, and its source, destination, services, action, tracking and installation settings were examined. Additional columns, including Hits, Track and Install On, were enabled to provide more information about rule activity and enforcement.

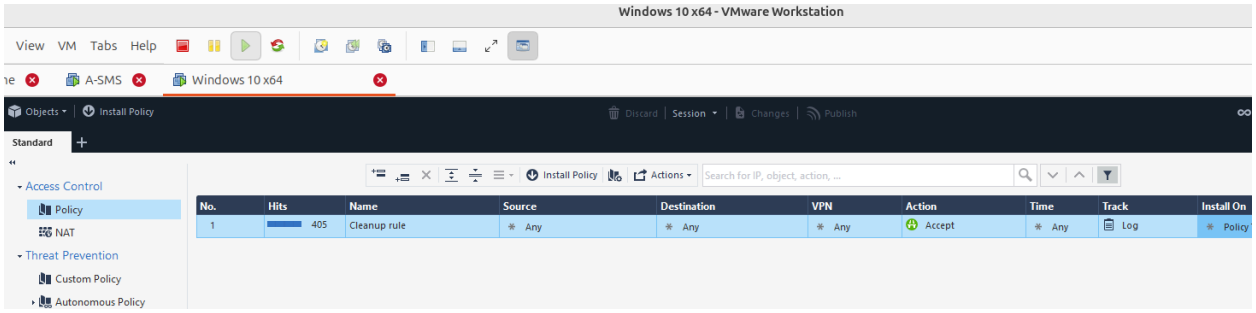
Task 1 - Evidence 1: Security Policies View



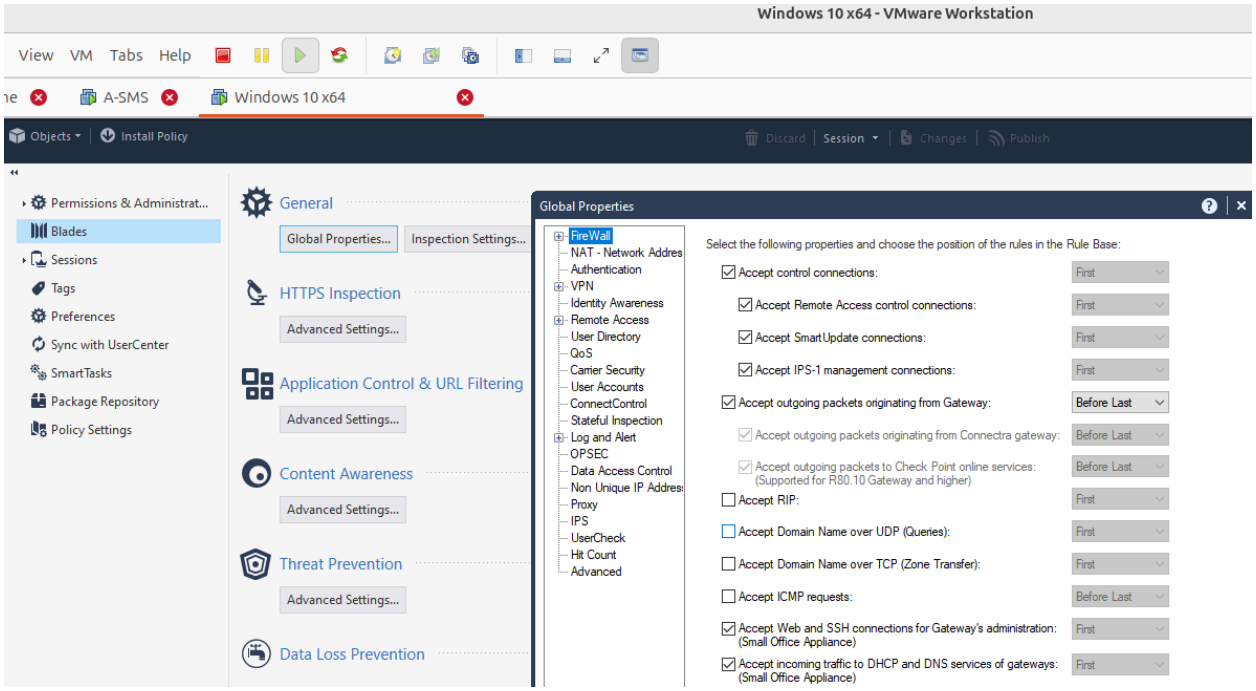
Task 1 - Cleanup Rule



Task 1 - Evidence 3: Custom Columns: added new column and hide the services & application menu

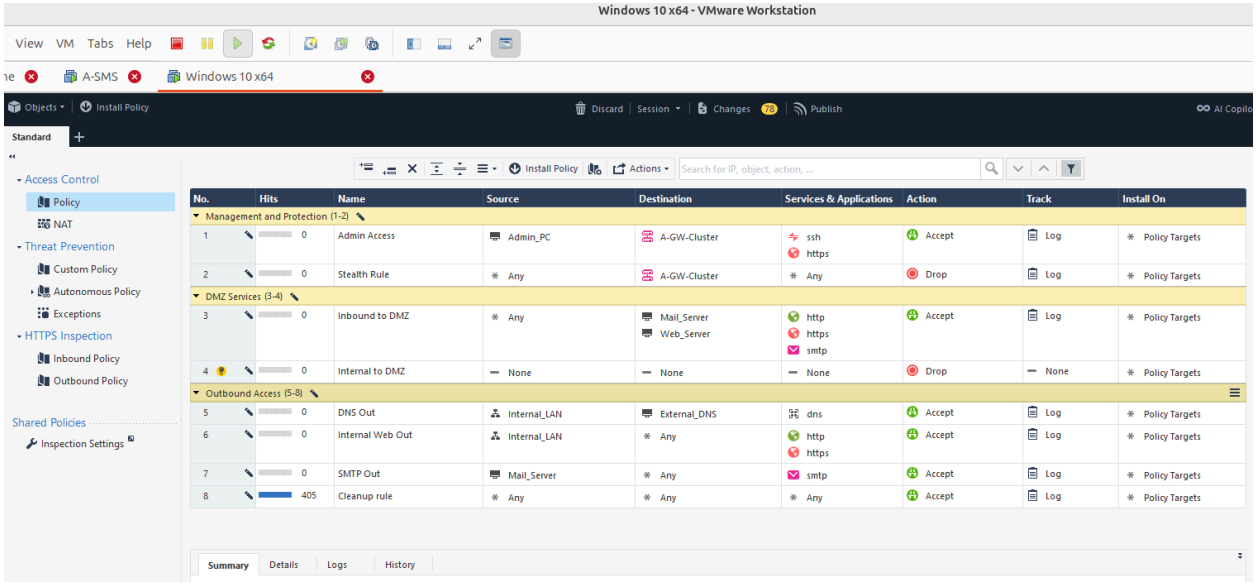


Firewall, VPN , Remote Access

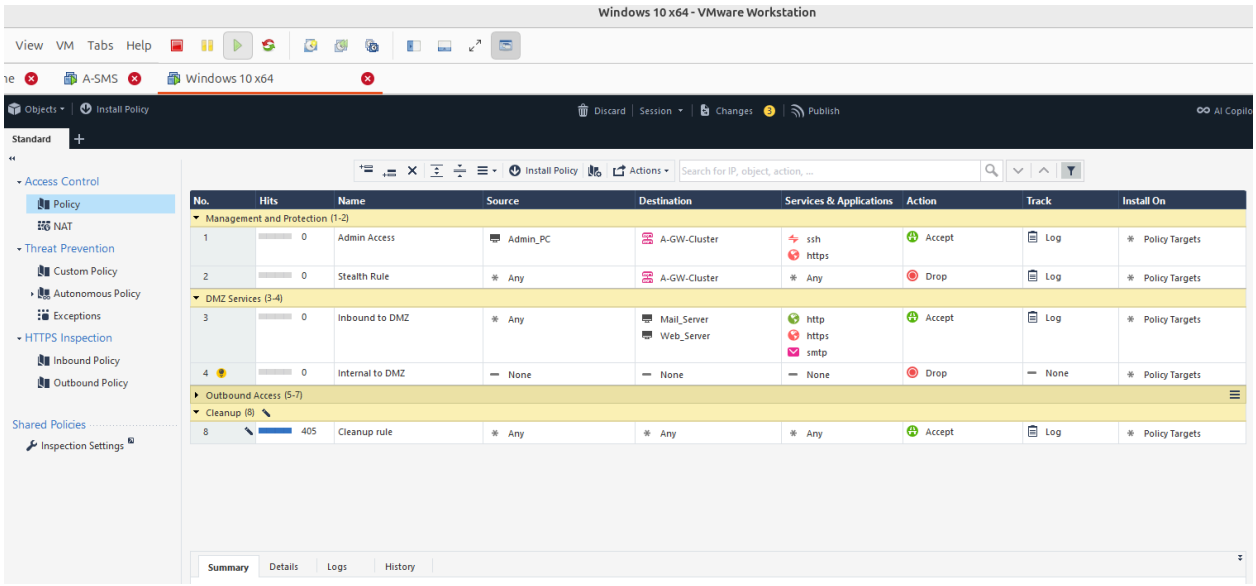


Configured Access Control rulebase for Orion Logistics showing administrative access, gateway protection, DMZ services, outbound access and cleanup controls

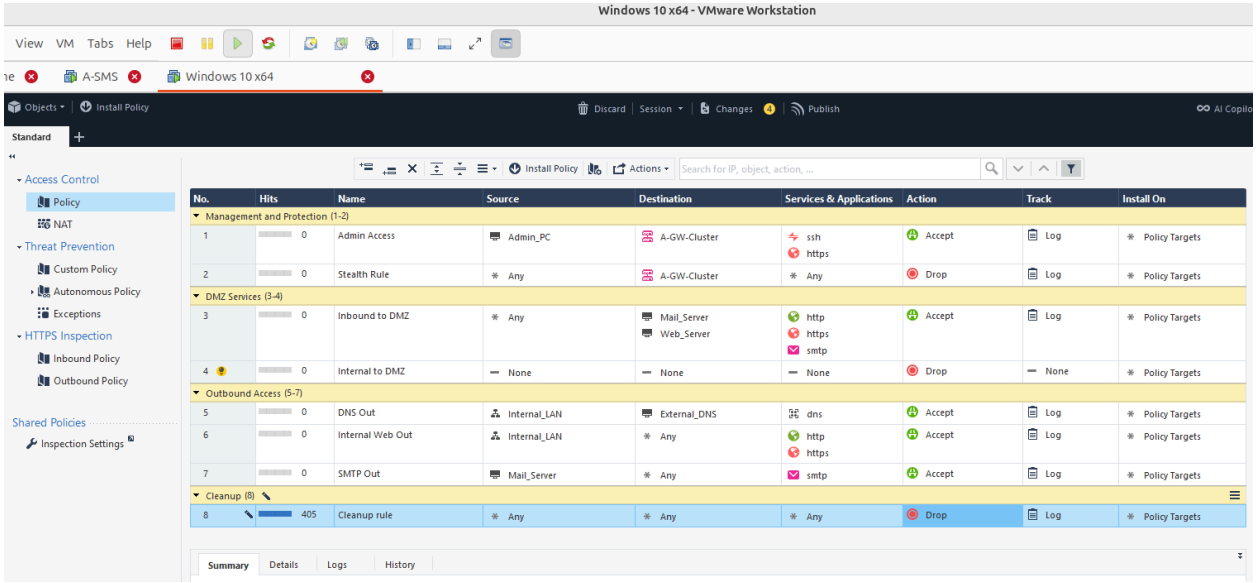
Access Control rulebase organised into Management and Protection, DMZ Services, Outbound Access and Cleanup sections.



Outbount Colapse: Access Control rulebase organised into Management and Protection, DMZ Services, Outbound Access and Cleanup sections.



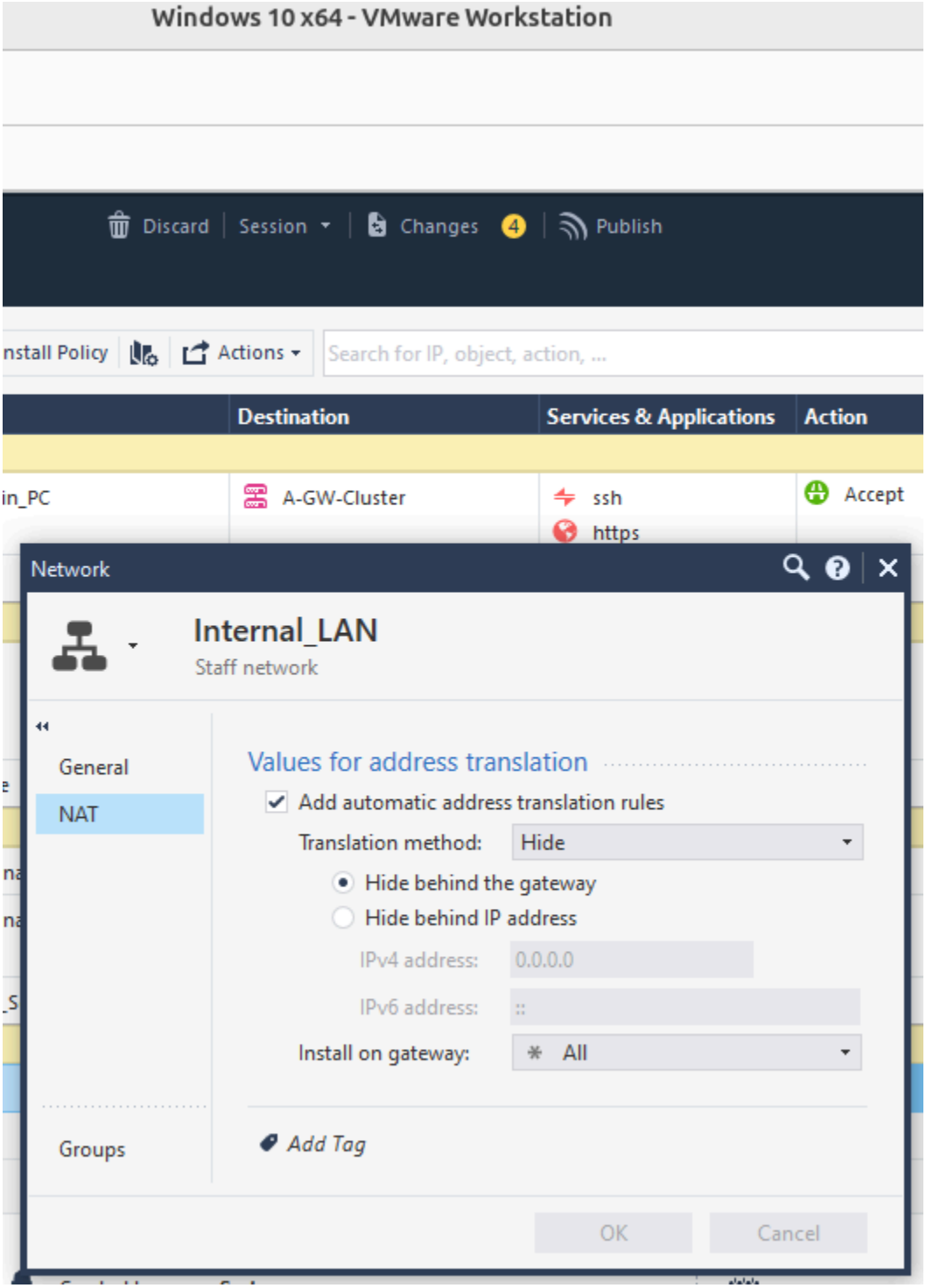
Outbound expanded: Demonstration of section collapse functionality in Check Point SmartConsole.



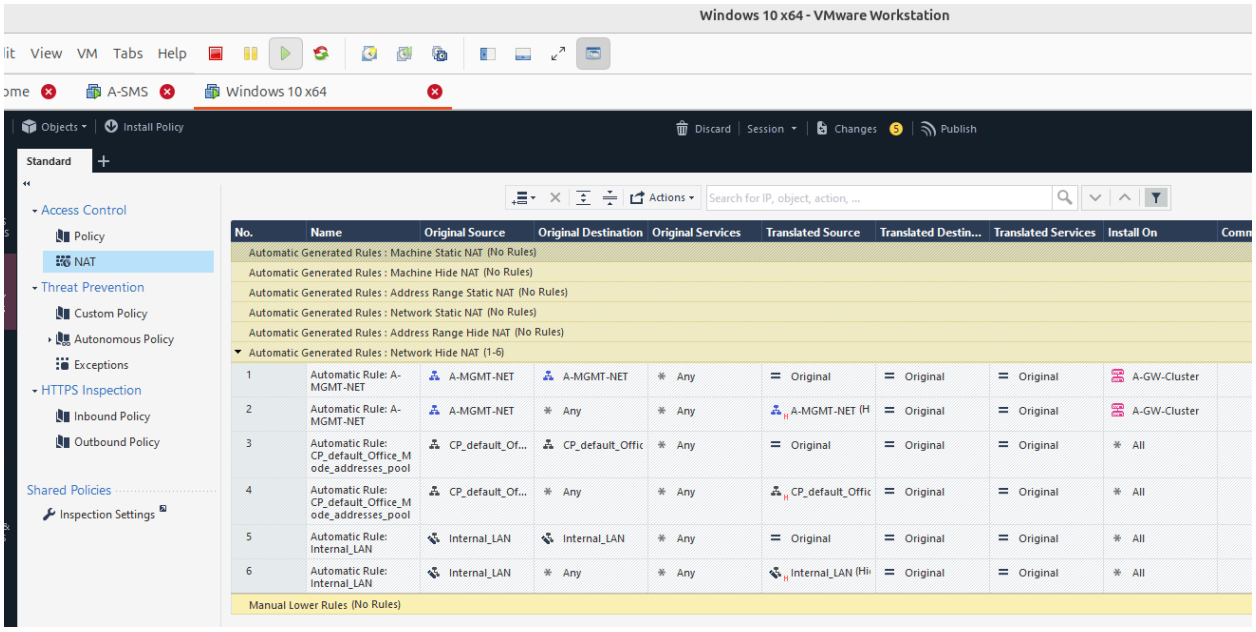
The Access Control rulebase was organised into four logical sections: Management and Protection, DMZ Services, Outbound Access and Cleanup. This arrangement improves readability and makes policy administration easier by grouping related security rules together.

The sections are visual organisational tools. They do not change the order in which firewall rules are evaluated and are not sent to the Security Gateway as enforcement rules. The actual rule sequence determines how traffic is evaluated.

Automatic Hide NAT configuration on the Internal_LAN network object.



Automatically generated Hide NAT rule for the Internal_LAN network.



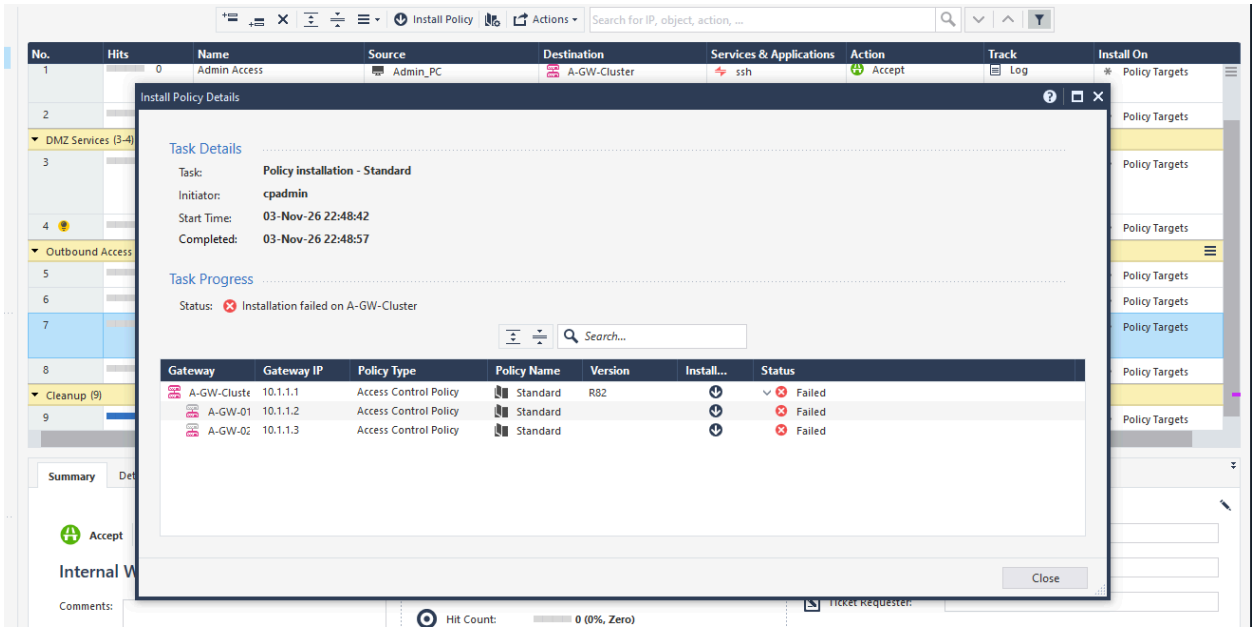
Hide NAT was configured on the Internal_LAN network object by enabling automatic address translation and selecting the Hide translation method. The network was configured to hide behind the Security Gateway.

An automatic NAT rule is generated from the NAT settings of the network object, whereas a manual NAT rule is created and configured directly by the administrator in the NAT rulebase.

Many-to-one translation means that multiple internal devices with different private IP addresses can share one translated IP address when communicating with external networks. The Gateway uses port translation to distinguish individual connections and maintain their return traffic correctly.

This configuration helps conserve public IP addresses and prevents internal private IP addresses from being directly exposed to external networks.

SmartConsole policy installation attempt showing failed installation on A-GW-Cluster and its associated gateway members because the gateway devices were unavailable.



Task 7: Rule Order Experiment and Installation History – Lab Limitation

The temporary blocking rule, TEMP Block Internal, was prepared to demonstrate the effect of firewall rule ordering. However, policy installation could not be completed because the Security Gateway devices were powered off.

The available laboratory resources were limited to the A-SMS management server and Windows 10 virtual machine due to the host computer's hardware capacity (16 GB RAM and 256 GB SSD). SmartConsole reported failed installation attempts on A-GW-Cluster (10.1.1.1), A-GW-01 (10.1.1.2) and A-GW-02 (10.1.1.3).

Consequently, actual traffic enforcement and corresponding gateway log verification could not be performed. The rule-order experiment was documented at the management configuration level, and the installation failure was retained as evidence rather than reporting an unverified successful result.

PART B

SECTION B1

Q1. Answer> B – Created in SmartConsole, stored on the Security Management Server and enforced by Security Gateways.
Q2. Answer> A – Positive Control Model (whitelist).
Q3. Answer> C – As early as possible, typically immediately after the Management rules.
Q4. Answer> D – Sections are visual divisions only and do not affect enforcement order.
Q5. Answer> B – Many-to-one.

SECTION B2

Q6. Distinguish between Explicit rules and Implied rules. State where Implied rules are configured and where they can be placed. What is the Implicit Cleanup Action?

Answer:

Explicit rules are security rules created and managed by the administrator in the Access Control rulebase. They define which network traffic is allowed or blocked based on parameters such as source, destination, services and applications.

Implied rules are predefined rules provided by Check Point to permit essential system communications, such as connections required for policy installation and communication between Security Gateways and the Management Server. They are configured through Global Properties in SmartConsole and can be positioned as First, Last or Before Last in the rulebase.

The Implicit Cleanup Action is the automatic final action applied to traffic that does not match any explicit or implied rule. Its default behaviour is to drop unmatched traffic, providing a final security control. Unlike an explicit Cleanup rule, it is not displayed in the rulebase.

Q7. Explain why rule order matters in a rulebase. Give one example from your own rulebase.

Answer:

Rule order matters because Check Point evaluates firewall rules sequentially, from top to bottom, and applies the first matching rule. Therefore, specific rules should be placed before general rules to ensure that the intended security decisions are enforced.

For example, in my Orion Logistics rulebase, the Admin Access rule allows the administrator's workstation to access the Security Gateway through SSH and HTTPS. This rule must appear before the Stealth Rule, which blocks other traffic directed to the Gateway. If the Stealth Rule were placed above Admin Access, legitimate administrative connections could be blocked before reaching the intended Allow rule.

Proper rule ordering therefore prevents unintended access and ensures that legitimate traffic is not unnecessarily denied.

Q8. Explain the difference between saving and publishing changes in SmartConsole. What happens to unpublished changes when you install a policy, and why do many organisations publish only during a change window?

Answer:

Saving changes in SmartConsole happens automatically as an administrator works within a session. These changes remain private to that administrator until they are published.

Publishing commits the changes to the Security Management Server's management database, making them available to other administrators and eligible for policy installation.

Unpublished changes are not included when a policy is installed. Only published changes are available for installation on the Security Gateway.

Many organisations publish changes during an approved change window to ensure that modifications are reviewed, tested and authorised before they become part of the organisation's security configuration. This reduces the risk of accidental policy changes, service disruption and unauthorised access.

Q9. Compare Global Properties with Policy Packages. Give one example of what each controls and say how you open each in SmartConsole.

Answer:

Global Properties are settings that control general Security Management and Security Gateway behaviour across the environment. For example, they can control firewall-related settings and the configuration of implied rules.

Policy Packages are collections of security policies and layers that administrators manage and install on selected Security Gateways. They determine the specific security rules enforced by a Gateway.

In SmartConsole, Global Properties can be accessed through Menu → Global Properties, while Policy Packages can be accessed through Security Policies → Manage policies and layers.

The main difference is that Global Properties control general security configuration, whereas Policy Packages contain the actual security policies applied to network devices.

Q10. Name the four predefined Security Zones. State two rules about assigning gateway interfaces to zones, and one benefit of using zone-based rules.

Answer:

The four predefined Check Point Security Zones are:

1. InternalZone – Represents trusted company networks containing internal resources.
2. ExternalZone – Represents untrusted networks, such as the Internet.
3. DMZZone – Represents the demilitarised zone containing publicly accessible company servers.
4. WirelessZone – Represents networks accessed through wireless connections.

Two important rules for assigning interfaces are:

- A Security Gateway interface must be assigned to an appropriate Security Zone before that zone can be used effectively in Access Control rules.

- The zone assignment should reflect the actual network connected to the interface, using its topology settings. For example, an interface connected to the Internet should be assigned to ExternalZone.

One major benefit of zone-based rules is that administrators can create simpler, reusable policies to control communication between network segments without having to specify every individual network address.

SECTION B3

Q11. A new administrator inserts a rule with Source Any, Destination Any, Services Any and Action Accept above the Stealth rule. Describe two security problems this causes, explain how to correct it, and name the type of rule that is allowed to sit above the Stealth rule and why.

Answer:

Placing an unrestricted Accept rule above the Stealth Rule creates serious security risks.

First, it allows traffic from any source to any destination using any service, potentially giving unauthorised users access to protected network resources.

Second, it defeats the purpose of the Positive Control Model because traffic is allowed without checking whether it is specifically required. It may also expose the Security Gateway and internal services to unnecessary access.

To correct this problem, the administrator should remove the unrestricted rule and replace it with specific rules that define approved sources, destinations and services. The Stealth Rule should remain near the top of the rulebase to protect the Security Gateway.

The only appropriate rules to place above the Stealth Rule are Management rules. These permit authorised administrators and essential management systems to communicate with the Security Gateway for administration and management purposes.

Q12. After Hide NAT is configured for Internal_LAN, staff report they cannot reach the Internet.

(a) State two benefits of NAT.

Answer:

1. Public IP address conservation: Hide NAT allows multiple internal devices to share one public IP address when accessing external networks, reducing the need for multiple public IP addresses.

2. Internal network address protection: NAT hides the private IP addresses of internal devices from external networks, making the internal addressing structure less visible.

(b) Describe three checks you would make to find the fault, and explain why each one matters. (3 marks)

Answer:

1. Verify the NAT configuration: I would check the Internal_LAN object's NAT settings to confirm that automatic Hide NAT is enabled and that the network is correctly hidden behind the Security Gateway. An incorrect translation configuration may prevent internal devices from communicating externally.

2. Check the Access Control policy: I would inspect the outbound rules to confirm that Internal_LAN is permitted to access the required external destinations and services, such as HTTP and HTTPS. A Drop rule or incorrect rule order could block the traffic even when NAT is correctly configured.

3. Check the Gateway's routing and connectivity: I would inspect the Gateway's external interface, default route and DNS configuration. The Gateway must have a working route to the Internet, while a functioning DNS resolver is necessary for users to access websites using domain names.

These checks help identify whether the problem originates from NAT translation, firewall policy enforcement or external network connectivity.